Computational Lab 7: PDAC Knowledge Graph Integration and Interpretation

A Neo4j knowledge graph was built to connect PDAC multi-omics evidence from Module 2, tractability priors from Module 3 and pharmacogenomic variants from Module 5. The graph supports drug–gene queries, candidate prioritization and interpretation of shared disease features. The committed run contained 80 nodes and 101 relationships. No patient layer was included, so patient stratification was not performed.

Genes were standardized to HGNC symbols, with Ensembl IDs retained as properties. Repeated references to CFTR and CEL were merged into single nodes. Six node kinds were included: genes, diseases, phenotypes, pathways, drugs and variants. Phenotypes connect diseases through shared clinical features; pathways provide mechanistic context; variants connect earlier pharmacogenomic work to drug handling.

Node kind	Gene	Disease	Phenotype	Pathway	Drug	Variant
Count	30	4	6	15	15	10

Table 1. Node counts in the committed graph. All 30 gene nodes had unique Ensembl IDs.

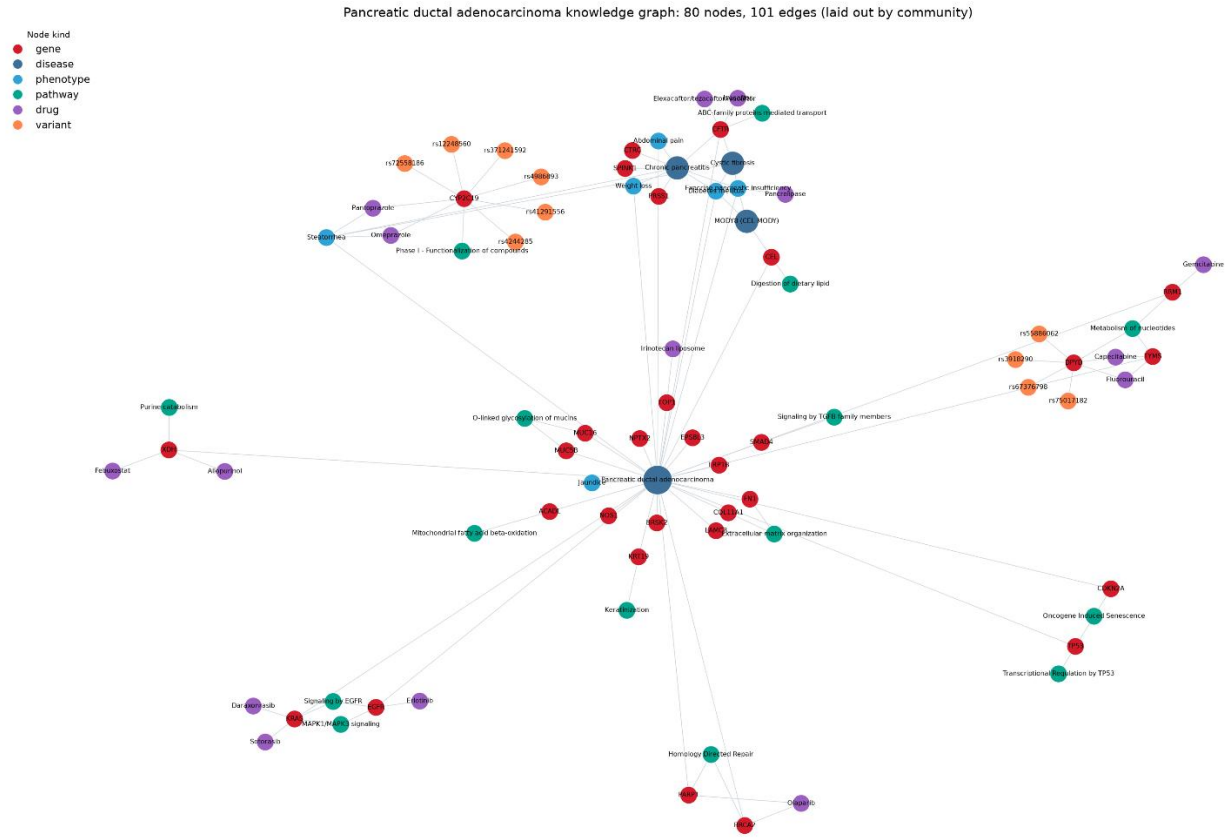


Figure 1. Overview of the committed graph, colored by node kind and arranged by community. The separate full-resolution PNG supports inspection of individual labels. A complete evidence path is presented on page 3.

Eight relationship types were used: ASSOCIATED_WITH (31), PARTICIPATES_IN (24), HAS_PHENOTYPE (16), TARGETS (12), OCCURS_IN (10), METABOLIZED_BY (4), TREATS (3) and PREDICTS_RESPONSE_TO (1). Every relationship carried a source and a confidence value, and the notebook confirms that none is missing either. Confidence was assigned on a four-band scale: 1.0 curated and definitive, such as an approved drug mechanism or a Mendelian gene–disease link, with 0.8–0.9 curated annotation, such as Reactome membership, an HPO phenotype or a ClinVar assertion with at least two review stars, 0.5–0.7 moderate, such as investigational drugs or conflicting and low-star ClinVar records, and 0.2–0.4 weak, such as pharmacology-only links, adjunctive therapy and provisional computational calls. These values represent heuristic evidence-strength scores, not probabilities. Module 2 association scores were retained as prioritization evidence rather than interpreted as calibrated certainty.

Queries and candidate prioritization

The opening query returned 12 drug–gene–PDAC rows. It included established treatment mechanisms and possible repurposing connections, such as gout drugs linked to XDH and cystic fibrosis modulators linked to CFTR. A returned path establishes a connection in the curated graph, and it does not establish that the drug treats PDAC. The shared-symptom extension traversed four relationships and returned four grouped rows: two CFTR modulators connected through cystic fibrosis and chronic pancreatitis.

Ranking method and committed results

Unweighted directed PageRank was computed in NetworkX and combined with earlier evidence using: fused score = 0.4 × gene PageRank percentile + 0.4 × Module 2 score + 0.2 × druggability. The Module 3 prior was used where available, otherwise if not, then a drug-status proxy was assigned. These components use different evidence and should not be interpreted as a probability of biomarker validity.

Rank	Gene	Fused Score	PageRank	M2 Score	Tractability Proxy
1	KRAS	0.885	0.0164	0.8113	1.0000 (drug status)
2	XDH	0.841	0.0164	0.9515	0.5035 (Module 3)
3	CFTR	0.760	0.0164	0.9781	0.0444 (Module 3)
4	CEL	0.668	0.0061	0.9620	0.7163 (Module 3)
5	MUC5B	0.626	0.0061	0.9419	0.5458 (Module 3)

Table 2. Top five candidates in the committed notebook and ranking CSV, after the drug-status correction described below.

KRAS provides a biologically recognizable result, its mutation frequency was 0.9619 in the integrated cohort, while its modest expression changes placed it only 69th in Module 2. It reaches first position because the graph supplies what the expression data did not, that is the pathway membership and two targeting drugs, and not because the omics evidence is strong. XDH is a non-obvious candidate.

Its high Module 2 score, tractability prior and incoming drug annotations explain its prominence, but no PDAC dependency or therapeutic benefit was demonstrated, and its approved drugs treat gout. CFTR and CEL were decreased in bulk tumor tissue, loss of normal pancreatic cell populations is one possible explanation requiring cell-specific validation.

TP53, BRCA2 and SMAD4 ranked 25th, 24th and 26th in the committed table. These low positions do not imply weak biological importance. The composite ranking depends on expression-based evidence, available tractability priors and the curated topology. Expression changes alone do not capture every mutation-driven mechanism.

Drug status correction and interpretation limits

Daraxonrasib was initially annotated as investigational, reflecting its trial status when the drug layer was curated. The FDA approved it on 26 August 2026 for adults with metastatic pancreatic adenocarcinoma who have received at least one prior systemic therapy or who are not candidates for multiagent systemic therapy [1]. The annotation was corrected to approved and the notebook re-run, which raises the KRAS tractability proxy from 0.5 to 1.0 under the existing rule and moves KRAS from second to first with a fused score of 0.885. Table 2, the exported ranking CSV and the figure all reflect the corrected run. It is worth recording that a drug-status annotation is a dated claim, and a ranking that weights druggability inherits whatever date its annotations were curated on.

PageRank follows incoming links, and outgoing pathway links do not automatically raise a gene's score. KRAS, XDH and CFTR tied on PageRank, while CYP2C19 and DPYD had the highest gene centrality because of incoming variant and metabolism links. The latter were excluded from PDAC biomarker ranking, because neither carries an association edge to the disease. KRAS led all four recorded weight tests after the correction, and XDH, CFTR and CEL remained in the top five throughout, although their order changed. Centrality and tractability do not establish diagnostic or prognostic biomarker performance.

A path based hypothesis and one community

Shared pancreatic dysfunction hypothesis

One full path was returned. Each row below is written in the direction the path is traversed from ivacaftor to PDAC, and the final relationship is stored as PDAC to phenotype and is therefore followed in reverse.

Step	Relationship and evidence	Reaches
Ivacaftor	TARGETS; mechanism annotation; score 1.0	CFTR
CFTR	ASSOCIATED_WITH; Mendelian association; score 1.0	Cystic fibrosis
Cystic fibrosis	HAS_PHENOTYPE; phenotype annotation; score 0.8	Exocrine pancreatic insufficiency
Exocrine pancreatic insufficiency	HAS_PHENOTYPE (traversed in reverse); clinical presentation; score 0.8	PDAC

Table 3. The four-edge path connects ivacaftor to PDAC through a shared phenotype. It generates a hypothesis, rather than evidence of efficacy.

The path suggests testing whether a subset of PDAC-associated pancreatic dysfunction involves a drug-responsive CFTR defect. CFTR was decreased in the supplied bulk data (RNA log2FC −2.438, protein −1.819). This finding cannot distinguish loss of ductal cells from reduced expression or impaired function within surviving cells. PDAC and cystic fibrosis also share a CFTR node in this graph, so the phenotype is not their only connection.

Testing would begin with cell-specific expression and protein-localization measurements in residual pancreatic ductal epithelium. Genotyping and functional transport assays would then establish whether CFTR is present and responsive to a modulator, and ivacaftor could be tested in patient-derived ductal organoids against matched controls using chloride or bicarbonate transport or swelling assays. Its established use depends on responsive CFTR variants and available channel protein [2], reduced expression alone does not establish responsiveness. The hypothesis would be weakened if dysfunction were explained mainly by irreversible obstruction, tissue destruction or absent channel protein, or if organoids showed no functional response. Any potential role would initially concern supportive pancreatic function, not an anticancer effect. A low computational tractability prior would not override established drug–target evidence.

Community interpretation

Greedy modularity clustering of the undirected graph identified eight communities with modularity 0.626. Community 1 contained 17 nodes, having three diseases, five genes, three drugs, two pathways and four phenotypes. CFTR, CEL, PRSS1, SPINK1 and CTRC were grouped with cystic fibrosis, chronic pancreatitis, MODY8 and pancreatic dysfunction phenotypes. This provides an interpretable exocrine-pancreas module.

The grouping suggests that part of the bulk PDAC signal may reflect changes in normal pancreatic function or cell composition, although that explanation was not tested. Community structure partly reflects the manually selected edges and is not independent biological validation or a discovered patient subtype. Other communities captured fluoropyrimidine pharmacogenomics, CYP2C19/PPI metabolism, KRAS–EGFR signaling, DNA repair and purine catabolism, all these partitions organize the selected evidence rather than prove distinct mechanisms.

Evidence limits and responsible use

Weak edges and their consequences

Pharmacology-only links from TYMS, RRM1, TOP1, EGFR and PARP1 to PDAC were assigned low scores because they indicate treatment relevance rather than independently measured disease association. PPI–steatorrhea links describe adjunctive supportive care and require context. The provisional CYP2C19 VUS annotation is computational and must not be presented as a clinical reclassification.

OCCURS_IN records variant location, which should be supported by mapping evidence separately from ClinVar significance or predicted functional effects. ClinVar review status and CPIC

pharmacogene function answer different questions; disagreement should be retained with its condition and source. A low CADD score does not exclude a pharmacogenomic effect. Patient-level interpretation would require genotype, zygosity, allele context and relevant clinical factors [3].

The BRCA2–olaparib relationship also requires qualification: a gene node alone is not a response biomarker. The labelled PDAC maintenance setting concerns qualifying germline BRCA mutations, metastatic disease and absence of progression after at least 16 weeks of first-line platinum treatment [4]. Drug targeting, approval status and PDAC indication should be stored as distinct annotations, as the daraxonrasib correction above illustrates, collapsing them into one status field makes the ranking sensitive to a single dated label.

Better data and verification

The graph inherits unmatched-normal comparisons, incomplete multi-omics coverage and manual annotation choices. Additional literature or disease-association evidence could change the ranking, but the direction and size of that change were not tested. External PDAC cohorts, cell-specific measurements and perturbation experiments would be needed to evaluate the candidates. Real patient genotypes with treatment and toxicity outcomes would be needed for patient stratification or clinical validation.

No patient stratification or independent clinical validation was performed for this report. The committed analysis supports hypothesis generation. It does not establish that the ranked genes are clinically useful biomarkers or that connected drugs benefit PDAC patients.

AI Use Disclosure

Claude (Anthropic, 2026) was used to structure the report and to mitigate errors. All content and analysis are my own. Anthropic. (2026). Claude (Opus 4.5) [Large language model]. https://claude.ai

References

1. FDA. FDA approves daraxonrasib for metastatic pancreatic adenocarcinoma. 26 August 2026. fda.gov/drugs/resources-information-approved-drugs/fda-approves-daraxonrasib-metastatic-pancreatic-adenocarcinoma.
2. FDA. Kalydeco prescribing information. 2025. Accessdata.fda.gov/grugsatfda_docs/label/2025/203188s042,207925s020lbl.pdf.
3. Amstutz U et al. CPIC guideline for DPYD genotype and fluoropyrimidine dosing: 2017 update. Clin Pharmacol Ther. 2018;103:210–216. doi:10.1002/cpt.911.
4. DailyMed. Lynparza prescribing information. Pancreatic adenocarcinoma indication and POLO study. dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=741ff3e3-dc1a-45a6-84e5-2481b27131aa.
5. Project inputs: targets_pdac.csv, targets_pdac_w3.csv, ensembl_symbol_mapping.csv, DPYD.tsv, CYP2C19.tsv, pgx_classified_CYP2C19.csv. Curated layers: HPO, Reactome, ChEMBL/DrugBank, CPIC.